

SRI CHAITANYA IIT ACADEMY, INDIA

AP • TELANGANA • KARNATAKA • TAMILNADU • MAHARASHTRA • DELHI • RANCHI

PRAGYA G8 – ACHIEVERS | MATHEMATICS
INTEGRATED PAPER I (HELL LEVEL) | JEE MAIN PATTERN*Every question draws on at least three of: Surds, Logarithms, Sequences & Series, Quadratic Equations***SECTION – A**
(SINGLE CORRECT ANSWER TYPE)

- This section contains TWENTY (20) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is correct.
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks: +4 if the correct option is chosen;**Zero Marks: 0 if the question is unanswered;**Negative Marks: –1 in all other cases.*

1. Let $s = \sqrt{3-2\sqrt{2}} + \sqrt{5-2\sqrt{6}} + \sqrt{7-4\sqrt{3}} + \sqrt{9-4\sqrt{5}} + \sqrt{11-2\sqrt{30}}$. If s is a root of $x^2 + px + q = 0$ where $p, q \in \mathbb{Z}$, and p, q, r are in arithmetic progression, then r equals
(A) -7 (B) -12
(C) -3 (D) 12
2. Let S be the sum of all real values of x satisfying $(x^2 - 5x + 5)^{x^2 - 9x + 20} = 1$. If S is the n th term of the arithmetic progression $3, 7, 11, \dots$, then $\log_2(S + 1) + n$ equals
(A) 6 (B) 9
(C) 7 (D) 8
3. Let $x_1 > x_2$ be the real numbers satisfying $\log_{(\sqrt{3}+\sqrt{2})} x \cdot \log_{(\sqrt{3}-\sqrt{2})} x = -4$. Then $x_1 + x_2 + \sqrt{x_1 x_2}$ equals
(A) 9 (B) 12
(C) 10 (D) 11
4. Let N be the number of reals $x > 0, x \neq 1$, satisfying $\log_{(\sqrt{5}+2)} x + \log_{(\sqrt{5}-2)} x = 4$. If N is the first term of an arithmetic progression whose fifth term is 12 , then its common difference is
(A) 12 (B) 4
(C) 3 (D) 2
5. If $\log_2 x, \log_2(x + 2)$ and $\log_2(3x + 2)$ are in arithmetic progression, and x is the fourth term of a geometric progression whose first term is $\frac{1}{4}$, then its common ratio is

(A) $\frac{1}{2}$

(B) 4

(C) $\sqrt{2}$

(D) 2

6. Let S be the sum to infinity of the geometric progression $2, \sqrt{2}, 1, \frac{1}{\sqrt{2}}, \dots$. If S is a root of $x^2 + px + q = 0$ where $p, q \in \mathbb{Z}$, then $p + q$ equals

(A) -16

(B) 8

(C) 0

(D) -8

7. Let P be the product of all real values of x satisfying $x^{\log_{10} x} = 100x^2$. If $\log_{10} P, k$ and 8 are in arithmetic progression, then k equals

(A) 6

(B) $\frac{11}{2}$

(C) 4

(D) 5

8. Let $\alpha > \beta$ be the roots of $x^2 - 2x - 1 = 0$. If $\alpha^3 - \beta^3, 20\sqrt{2}$ and k are in arithmetic progression, then k equals

(A) $24\sqrt{2}$

(B) $32\sqrt{2}$

(C) $28\sqrt{2}$

(D) $30\sqrt{2}$

9. Suppose $(\log_2 x)^2 - (k+2)\log_2 x + (2k+1) = 0$ has both of its roots (as values of $\log_2 x$) integers, not necessarily distinct. The two admissible values of k , in increasing order, are the first and third terms of an arithmetic progression. Its second term is

(A) 4

(B) 1

(C) 2

(D) 0

10. Three distinct non-zero reals a, b, c are in geometric progression and $a, 2b, 3c$ are in arithmetic progression. If $a = \sqrt{2}$, then the sum to infinity of the geometric progression is

(A) $\frac{3\sqrt{2}}{2}$

(B) $\frac{\sqrt{2}}{2}$

(C) $\frac{2\sqrt{2}}{3}$

(D) $3\sqrt{2}$

11. Let x_1 and x_2 be the roots of $\log_3 x + \log_x 3 = \frac{5}{2}$. If $\log_3 (x_1 x_2), k$ and 10 are in arithmetic progression, then k equals

(A) $\frac{21}{4}$

(B) $\frac{25}{4}$

(C) $\frac{27}{4}$

(D) $\frac{23}{4}$

12. Let $A = \sqrt{2 + \sqrt{2 + \sqrt{2 + \cdots \infty}}}$ and $B = \sqrt{6 + \sqrt{6 + \sqrt{6 + \cdots \infty}}}$. If A, G, B are in geometric progression with $G > 0$, then $\log_2(A + B + G^2 - 3)$ equals
- (A) 4 (B) 3
(C) 2 (D) $\log_2 11$
13. Let $\alpha = \sqrt{19 + 6\sqrt{2}}$ and $\beta = \sqrt{19 - 6\sqrt{2}}$. If $\alpha - \beta, 4$ and k are in geometric progression, then k equals
- (A) 8 (B) 12
(C) 6 (D) 10
14. Let N be the number of real solutions of $2 \log_3 x - \log_3(x + 6) = 1$ and let x_0 be their sum. Then $N + x_0$ equals
- (A) 7 (B) 9
(C) 5 (D) 6
15. Three reals $a < b < c$ are in arithmetic progression with $a + b + c = 15$ and $a^2 + b^2 + c^2 = 83$. If a, g, c are in geometric progression with $g > 0$, then g equals
- (A) $\sqrt{35}$ (B) $\sqrt{21}$
(C) $\sqrt{15}$ (D) 5
16. The number of integral values of x satisfying $\sqrt{x + 3 - 4\sqrt{x - 1}} + \sqrt{x + 8 - 6\sqrt{x - 1}} = 1$ is the third term of a geometric progression whose first term is $\frac{2}{3}$. Its positive common ratio is
- (A) $\frac{3}{2}$ (B) $\sqrt{6}$
(C) 3 (D) 2
17. Let the sum of all real solutions of $4^x - 3 \cdot 2^{x+1} + 8 = 0$ be the second term of an arithmetic progression whose first term is 1. Its fifth term is
- (A) 11 (B) 9
(C) 13 (D) 7
18. Let G and H be the geometric mean and the harmonic mean of $3 + 2\sqrt{2}$ and $3 - 2\sqrt{2}$. If $\frac{G}{H}, k, 27$ are in geometric progression with $k > 0$, then $\log_3 k$ equals
- (A) 2 (B) $\frac{3}{2}$
(C) 3 (D) 1
19. Let α, β be the roots of $x^2 - 4x + 2 = 0$. If $\log_2(\alpha\beta), \log_2(\alpha - \beta)^2$ and k are in arithmetic progression, then k equals

(A) 5

(B) 6

(C) 7

(D) 4

20. Let $x = \log_2 3 \cdot \log_3 4 \cdot \log_4 5 \cdots \log_{63} 64$. If x, y and $\sqrt{x^2 + 13}$ are in geometric progression with $y > 0$, then y^2 equals

(A) 36

(B) $\sqrt{42}$

(C) 42

(D) 49

SECTION – B
(NUMERICAL VALUE ANSWER TYPE)

- This section contains FIVE (05) questions.
- The answer to each question is a NUMERICAL VALUE. Enter the value in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 if the correct option is chosen;
Zero Marks: 0 if the question is unanswered;
Negative Marks: –1 in all other cases.

21. Given $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, let d be the number of digits in $(\sqrt{18})^{30}$ and let z be the number of zeros after the decimal point and before the first significant digit of $(\sqrt{18})^{-30}$. If $d + z$ is the n th term of the arithmetic progression $1, 4, 7, \dots$, then n is

22. Let $\alpha > \beta$ be the roots of $x^2 - 6x + 4 = 0$. If $\alpha^3 - \beta^3 = m\sqrt{5}$ and m is the n th term of the geometric progression $1, 2, 4, 8, \dots$, then n is

23. Three positive reals a, b, c are in geometric progression with $a + b + c = 13$ and $a^2 + b^2 + c^2 = 91$. If M is the largest of them, then $\log_3 M$ is

24. Let $\log_{12} 18 = a$ and $\log_{24} 54 = b$, and let $m = ab + 5(a - b)$. If $m, m + d, m + 2d$ are in arithmetic progression with sum 12, then d is

25. The third and seventh terms of an arithmetic progression are $\sqrt{2}$ and $5\sqrt{2}$. If these two terms are the roots of $x^2 + px + q = 0$ and the fifteenth term of the progression is $k\sqrt{2}$, then $k + q$ is
